

GRAMSEVA

ಗ್ರಾಮ ಸೇವಾ: ಗ್ರಾಮ ಮೂಲಸೌಕರ್ಯ ನಿರ್ವಹಣಾ
ವೇದಿಕೆ

Proactive Village Infrastructure Monitoring, Transparent Rule-Based
Prioritization, and Citizen-Verified Resolution Platform

OFFICIAL TECHNICAL PROJECT SPECIFICATION & ARCHITECTURE REPORT

This document provides comprehensive technical, architectural, and operational documentation for the **GramSeva** platform. Designed specifically for Gram Panchayats in Karnataka, GramSeva re-engineers rural governance from reactive complaint-handling into proactive preventative maintenance with deterministic algorithmic scoring, multi-tier escalation, and citizen verification loops.

PILOT GRAM PANCHAYAT

Honnur Gram Panchayat (ಹೊನ್ನೂರು)

Nagamangala Taluk, Mandya District, Karnataka

SYSTEM VERSION & RELEASE

v1.0.0 Production Build (Node 25 / React 18)

CommonJS Backend + Vite SPA Frontend

OPERATING PARADIGM

Proactive Field Audits & Two-Stage Verification

Deterministic Priority Engine (Zero Black-Box ML)

DOCUMENTATION CLASSIFICATION

Full Architectural & Technical Manual

Date: September 2026 • Document ID: GS-2026-DOC-01

1. Executive Summary & Problem Formulation

1.1 The Rural Infrastructure Crisis in Panchayats

In India's rural local governance ecosystem (Panchayati Raj Institutions - PRIs), public infrastructure assets such as **drinking water purification plants (RO units), borewells, street lighting networks, open stormwater drainage canals, and village interior roads** represent the physical backbone of community health and livelihood. However, traditional municipal grievance platforms suffer from fundamental systemic failures:

- **100% Reactive Dependence:** Faults remain undetected until a resident experiences severe distress and navigates bureaucratic hurdles to register a complaint. If a street fixture or water pump breaks in an elderly or disadvantaged neighborhood, it often remains neglected for months.
- **The "Ghost Closure" Epidemic:** Across legacy portals, work tickets are frequently closed administratively by department contractors without on-ground repairs having occurred, destroying citizen trust.
- **Chronic Asset Bleed:** Certain assets fail repeatedly (e.g., defective capacitor panels or submerged wiring), yet maintenance staff repeatedly apply superficial temporary fixes, consuming village budgets while the root cause remains unaddressed.
- **Opaque Prioritization:** Tickets are processed based on political favoritism or arbitrary queues rather than public safety urgency (e.g., a streetlamp failing outside a primary school vs. an isolated private lane).

1.2 The GramSeva Solution Paradigm

GramSeva (ಗ್ರಾಮ ಸೇವಾ) re-architects rural civic governance from *reactive complaint logging* to **proactive infrastructure reliability engineering**. Tested and piloted in **Honnur Gram Panchayat, Mandya District**, the platform introduces six foundational pillars:

100%

ASSET TAGGED REGISTRY

Weekly

PROACTIVE AUDITS

3-Tier

SLA ESCALATION

2-Stage

CITIZEN VERIFICATION

Core Hypothesis of GramSeva

By combining scheduled preventative inspection checklists with an auditable rule-based priority engine and a mandatory citizen verification loop, a Gram Panchayat can eliminate 60%+ of citizen grievances before they arise while ensuring 100% audit integrity for village maintenance funds.

1.3 Key Architectural Differentiators

DIMENSION	LEGACY GRIEVANCE REDRESSAL PORTALS	GRAMSEVA PROACTIVE GOVERNANCE PLATFORM
Discovery Mode	Purely reactive (waits for citizen complaint).	Proactive First: Scheduled digital inspection runs by linemen, auto-creating tickets prior to public escalation.
Prioritization	First-in, first-out (FIFO) or political influence.	Deterministic Scoring (0–100): Evaluates proximity to schools/PHCs, asset history, and safety hazards.
Closure Authority	Lineman/contractor marks "Resolved" unilaterally.	Dual Verification: "Resolved" is merely a verification request; issue is CLOSED only upon citizen confirmation.
Chronic Failures	Treated as disconnected random events.	Automated Recurrence Engine: Flags assets failing ≥ 3 times in 60 days, mandating overhaul over temporary repair.
Accountability	Single deadline with opaque extensions.	3-Tier Dynamic SLA Escalation: Auto-escalates from Lineman → Panchayat Development Officer (PDO) → Taluk Executive.

2. Pilot Jurisdiction Profile: Honnur Gram Panchayat

2.1 Geographic & Administrative Overview

The initial operational pilot of GramSeva is configured for **Honnur Gram Panchayat**, situated in Nagamangala Taluk of Mandya District, Karnataka (PIN: 571432). Honnur features a representative cross-section of rural infrastructure, spanning historical temple settlement zones, rural health hubs, active agricultural trade junctions, and newer peripheral community residential colonies.

PILOT ENTITY STRUCTURE

Honnur Gram Panchayat (ಹೊನ್ನೂರು ಗ್ರಾಮ ಪಂಚಾಯತಿ)

Taluk: Nagamangala | District: Mandya | State: Karnataka | Local Body Code: PRI-KAR-MAN-042

2.2 Ward Demographics & Infrastructure Focus

Honnur Gram Panchayat is partitioned into four primary administrative wards, each characterized by distinct civic infrastructure configurations:

WARD ID	WARD NAME	ELECTED REPRESENTATIVE	KEY PUBLIC INFRASTRUCTURE & SENSITIVITIES
w-1	Ward 1 - Kote & Temple Area	Somanna Gowda	Historical high-density settlement; ancient temple plaza, community hall (Kalyana Mantapa), stone drainage networks, heritage illumination poles.
w-2	Ward 2 - School & PHC Ward	Shivalingaiah	High-Sensitivity Zone: Government Higher Primary School, Primary Health Centre (PHC), Anganwadi Centre, main overhead water reservoir.
w-3	Ward 3 - Market & Bus Stop	Anitha Venkatesh	Commercial transport hub; KSRTC rural bus stop, weekly shandy/market stalls, public sanitations, multi-tap drinking water kiosk (RO plant).
w-4	Ward 4 - Colony & Lake Extension	Manjunatha K	Residential agricultural extension; lake bund roadway, irrigation outflow channels, decentralized borewells, peripheral streetlights.

2.3 Departmental Maintenance Cells

Maintenance operations are delegated across four dedicated operational departments within the Gram Panchayat administration, each governed by specific Service Level Agreements (SLAs):

DEPARTMENT CODE	DEPARTMENT NAME	OFFICER-IN-CHARGE	CRITICAL SLA	HIGH SLA	MED SLA
STREETLIGHT	Streetlight Maintenance Team	S. Nanjappa (Assistant Engineer, Electrical)	24 Hours	48 Hours	120 Hours
WATER_SUPPLY	Rural Water Supply & Sanitation (RWSS)	K. R. Vasanth (Junior Engineer, Water Works)	12 Hours	36 Hours	96 Hours
DRAINAGE_SANITATION	Gram Sanitation & Solid Waste Team	B. Puttegowda (Sanitation Officer)	18 Hours	48 Hours	72 Hours
ROADS_INFRA	Rural Infrastructure & PWD Cell	H. Manjunath (Section Officer, PWD)	48 Hours	96 Hours	168 Hours

LAYER	TECHNOLOGY / LIBRARY	ARCHITECTURAL RATIONALE & IMPLEMENTATION DETAILS
		mobile touch optimizations.
State & Localization	React Context API	Decoupled AuthContext, ThemeContext, and LanguageContext providing instant English/Kannada bilingual toggles.
Backend Engine	Node.js (v25+) + Express 5.2	High-concurrency asynchronous I/O, native JSON body parsing, structured modular controllers, robust error middleware.
Database Engine	SQLite Relational Database	Zero-configuration serverless database embedded in the filesystem (data/gramseva.db), supporting full ACID transactions and relational joins.
Security & Auth	JWT + Bcrypt.js	Stateless JSON Web Tokens with 7-day validity, role-based access control (CITIZEN, FIELD_STAFF, ADMIN), Bcrypt password hashing.
Media Pipeline	Multer 2.3 + Static Express	Multipart image file upload pipeline with MIME type verification and timestamped disk storage in /uploads.
Notifications	Nodemailer 10.0 + In-App Dispatcher	Dual-channel notification engine dispatching styled HTML transactional emails alongside persistent in-app notifications.

4. Proactive Inspection Engine & Asset Registry

4.1 The Public Asset Registry Model

Every public infrastructure asset across the village is cataloged within the assets relational table. Each entry encapsulates not only geographic and departmental ownership, but also operational risk levels and maintenance cadences:

- **Unique Asset Identifier:** E.g., SL-047 (Streetlight 47), WP-012 (Water RO Plant 12), DR-005 (Drainage Channel 5).
- **Risk Level Classification:** HIGH, MEDIUM, or LOW. Assets situated near schools or medical clinics are permanently designated as HIGH risk.
- **Inspection Frequency Cadence:** Standard default of 14 days (bi-weekly). Dynamically compressed to **7 days** if an asset is flagged for chronic recurrence.
- **Failure Counter:** Maintains a rolling tally of operational failures recorded across the asset's lifecycle.

4.2 Scheduled Field Inspection Runner

Rather than waiting for citizen complaints, GramSeva empowers field staff (linemen, water pump operators, sanitation workers) to conduct scheduled ward audits using the **Inspection Task Runner**.



4.3 The Proactive Detection Algorithm (Code Analysis)

When an inspector records a defect during a field audit, `handleInspectionItemResult()` in `server/services/proactiveDetector.js` executes the following deterministic sequence:

```
// Excerpt from server/services/proactiveDetector.js
1. Asset Status Update:
  - If result === 'NOT_WORKING' -> Update asset to 'NOT_WORKING' (or 'RECURRING_FAILURE' if flag is active).
  - Update asset.last_inspection_date to CURRENT_TIMESTAMP.

2. Duplicate Prevention Gate:
  - Queries existing unclosed issues for this specific asset_id.
  - If found: Links inspection as reaffirming evidence; prevents duplicate ticket clutter.

3. Automated Issue Synthesis:
  - Generates unique ID: GS-YYYY-XXXX.
  - Sets detection_source = 'SCHEDULED_INSPECTION'.
  - Calculates priority score & rationale via calculatePriority().
  - Calculates SLA deadline and assigns staff via routeAndAssign().
  - Inserts ticket in 'ASSIGNED' state directly (bypassing redundant verification).
  - Dispatches instant notifications to Admin and assigned Lineman.
```

5. Transparent Rule-Based Priority Engine

5.1 The Case Against "Black-Box" Machine Learning in Rural PRIs

Many modern civic platforms attempt to apply complex deep learning or opaque NLP models to prioritize complaints. In Indian rural local governance, this approach fails catastrophically:

- Panchayat Development Officers (PDOs) cannot explain to elected ward members why one complaint was prioritized over another.
- Training datasets in rural Kannada/English code-mixed dialects are noisy, introducing severe bias against illiterate or non-technical citizens.
- Black-box models cannot guarantee statutory SLA adherence for sensitive public installations (e.g., primary health centres or schools).

GramSeva implements an **Auditable, Deterministic Rule-Based Scoring Engine** (`server/services/priorityEngine.js`). The algorithm computes an exact integer score (0–100+) and generates an explicit human-readable rationale detailing every point added.

5.2 Priority Scoring Mathematical Breakdown

EVALUATION VECTOR	POINTS	CONDITION & TRIGGER CRITERIA
Baseline Weight	+20	Universal base score applied to every registered issue to ensure baseline visibility.
Sensitive Facility: Education	+30	Location text/landmarks matching: school, vidyalaya, shale. Prevents child hazards.
Sensitive Facility: Healthcare	+30	Location text/landmarks matching: phc, hospital, clinic, arogya. Protects medical clinics.
Sensitive Facility: Child Welfare	+25	Location text/landmarks matching: anganwadi. Prioritizes mother and child nutrition centers.
Transit & Commercial Density	+20	Location text/landmarks matching: bus stand, market, shandy. High pedestrian footfall.
Community Gathering Plaza	+15	Location text/landmarks matching: temple, kalyana mantapa. Public cultural hubs.
Essential Amenity: Water	+25	Category is WATER_POINT (drinking water RO plants, public taps, main borewells).
Sanitation & Flood Risk	+20	Category is DRAINAGE (open drain blockages, sewage overflow, public health risk).

EVALUATION VECTOR	POINTS	CONDITION & TRIGGER CRITERIA
Public Safety Hazard Keyword	+25	Descriptions matching: sparking, live wire, shock, contamination, burst, deep pothole.
Chronic Asset History	+25	Asset failure count ≥ 3 or flagged as <code>is_recurring_flag = 1</code> .
High-Risk Asset Designation	+15	Pre-designated high-risk public asset in master registry.
Aging / Duration Unresolved	+15 / +25	+15 if unresolved for ≥ 3 days; +25 if unresolved for ≥ 7 days. Prevents stale tickets.

5.3 Score-to-Priority Band Mapping

Once calculated, the cumulative score maps deterministically into four actionable operational bands:

CRITICAL SCORE ≥ 75 POINTS	HIGH SCORE 50 – 74 POINTS	MEDIUM SCORE 30 – 49 POINTS	LOW SCORE < 30 POINTS
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Audit Rationale String Example

"*+20 Baseline issue registration, +30 Near School / Child pedestrian zone, +20 Nighttime pedestrian visibility & safety, +25 Immediate public safety hazard [Score: 95 → CRITICAL]*"

6. Dynamic Escalation Matrix & Dual Verification

6.1 3-Tier Automated Escalation Hierarchy

In traditional governance systems, when a field worker misses a deadline, tickets simply languish indefinitely. GramSeva implements an **Automated 3-Tier SLA Escalation Engine** (server/services/escalationService.js). The service continuously calculates overdue hours against target completion dates:

ESCALATION TIER	BREACH THRESHOLD	RESPONSIBLE ADMINISTRATIVE OFFICER	ACTION & SYSTEM BEHAVIOR
LEVEL 1	> 0 Hours Overdue	Duty Lineman & Department Section Head	Ticket marked ESCALATED; reminder dispatched to field staff; highlighted in orange on Admin dashboard.
LEVEL 2	> 24 Hours Overdue	K. Shivakumar (Panchayat Development Officer - PDO)	Formal administrative escalation to PDO; emergency work order re-assignment authorized; flagged on Gram Panchayat review list.
LEVEL 3	> 72 Hours Overdue	Executive Officer (EO) (Taluk Panchayat, Nagamangala)	Taluk-level executive intervention; recorded in quarterly PRI performance audit; contractor penalty triggers logged.

6.2 The Complete 11-State Lifecycle Machine

Every issue navigates a strictly enforced state machine that prohibits administrative shortcuts:

```
[REPORTED] (Citizen or Community Observer files complaint)
|
v
[VERIFICATION_PENDING] ---> [VERIFIED] (Admin / Field staff confirms on ground)
|
v
[ASSIGNED] (Auto-routed to Dept & Staff with computed SLA)
|
+---> [ESCALATED] (Triggered if SLA target breached at any point)
|
v
[IN_PROGRESS] (Field staff clocks in on-site, logs initial repair notes)
|
v
[RESOLVED] (Field staff submits Mandatory AFTER-PHOTO & Completion Notes)
|
v
[CITIZEN_VERIFICATION] (Automated Notification sent to Citizen with Feedback link)
|
```

+-----+-----+	
v (Citizen confirms YES)	v (Citizen clicks NO / Unresolve
[CLOSED]	[REOPENED]
(Rating recorded, SLA marked complete)	(Bounces back to Lineman queue)

6.3 The Two-Stage Verification Loop & "Ghost Closure" Prevention

GramSeva Core Integrity Rule

Field staff and department contractors are physically prohibited from marking an issue as CLOSED. When work is finished, staff can only transition the status to RESOLVED, which requires uploading an *After-Repair Photo* and resolution notes. The system immediately shifts the issue to CITIZEN_VERIFICATION. The ticket is officially closed **ONLY** when the reporting resident confirms resolution or rates the work.

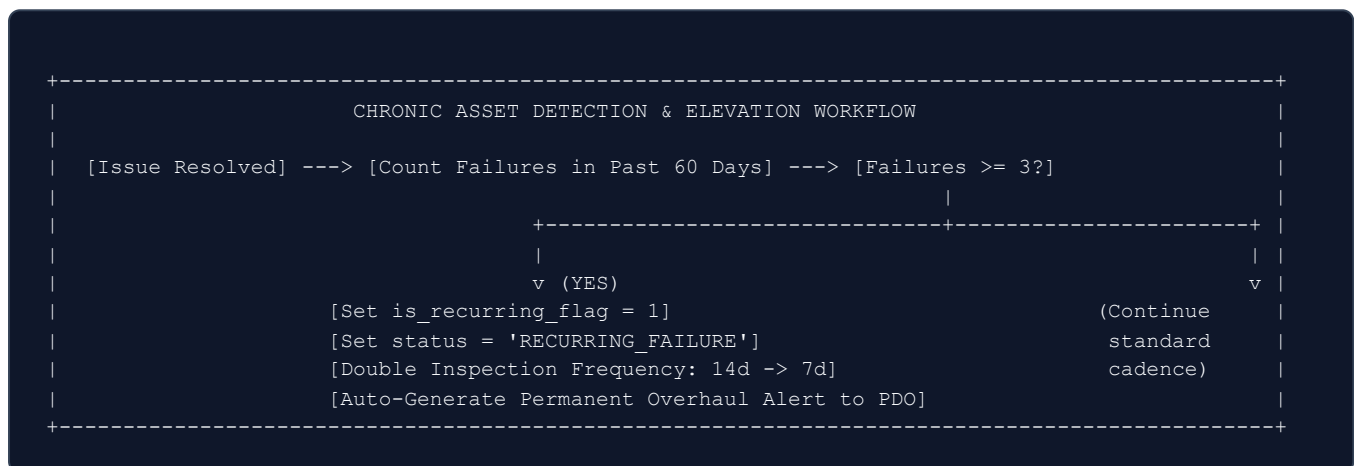


7. Chronic Asset Recurrence & Root-Cause Overhaul

7.1 Identifying the "Repair Money Pit"

A major leak in Gram Panchayat finances is repeated minor expenditure on dying infrastructure assets. For example, a village streetlight fixture with degraded underground insulation may short-circuit every 10 days. Linemen repeatedly replace fuses, billing the Panchayat for small repairs, while the underlying fixture remains dangerous and defective.

GramSeva's `checkAndFlagRecurringAsset()` algorithm monitors rolling 60-day failure histories across all public assets.



7.2 The 12-Dimensional Analytics Intelligence Suite

The GramSeva Analytics Engine (server/services/analyticsService.js) computes 12 distinct metric dimensions to provide the Panchayat Development Officer and Elected Ward Committee with complete operational visibility:

#	ANALYTICAL DIMENSION	GOVERNANCE PURPOSE & DECISION-MAKING IMPACT
1	Resolution Time by Category	Identifies which engineering cells (water vs electrical vs roads) face material or staffing bottlenecks.
2	6-Month Resolution Trend	Tracks month-over-month platform efficacy and resolution velocity over time.
3	Pending vs Resolved by Ward	Ensures equitable public spending and maintenance attention across all four village wards.
4	Category Distribution	Visualizes infrastructure breakdown volume (e.g., streetlights vs drinking water points).
5	Department SLA Compliance	Monitors average turnaround hours per department against statutory citizen charter benchmarks.

#	ANALYTICAL DIMENSION	GOVERNANCE PURPOSE & DECISION-MAKING IMPACT
6	Priority Distribution	Displays the ratio of Critical and High safety issues relative to routine Medium/Low tasks.
7	Chronic Recurring Assets Table	Lists top failing infrastructure assets requiring capital replacement in the next Gram Sabha budget.
8	Ward Vulnerability Hotspots	Highlights wards accumulating high concentrations of critical and high-priority infrastructure issues.
9	Citizen Feedback Ratings	Aggregates verified resident satisfaction distribution (1-Star to 5-Star ratings).
10	SLA Breach & Escalation Level	Analyzes ticket volume resolved within SLA vs those escalated to Level 1, Level 2, or Level 3.
11	Resolution Velocity Buckets	Groups resolved tickets into speed tiers: <24 hours, 1–3 days, 3–7 days, and >7 days.
12	Period Comparison	Compares current 30-day performance against previous 30-day performance for issues detected, resolved, and escalated.

8. Database Architecture & Relational Schema

8.1 Entity-Relationship Overview

GramSeva utilizes an ACID-compliant relational SQLite database with strict foreign key integrity enforced (`PRAGMA foreign_keys = ON;`). Below is the comprehensive schema definition of the 12 core tables:

TABLE NAME	PRIMARY KEY & FOREIGN KEYS	CORE ATTRIBUTES & CHECK CONSTRAINTS
villages	id (PK)	name, taluk, district, pin_code, gp_name, created_at
wards	id (PK), village_id (FK)	ward_number (INT), name, representative_name, created_at
users	id (PK), ward_id (FK)	name, email (UNIQUE), phone, password_hash, role (CITIZEN, FIELD_STAFF, ADMIN), designation, is_active
departments	id (PK), code (UNIQUE)	name, head_name, sla_hours_critical, sla_hours_high, sla_hours_medium, sla_hours_low
assets	id (PK), village_id (FK), ward_id (FK), department_id (FK)	asset_type (STREETLIGHT, WATER_POINT, PUBLIC_TOILET, ROAD, DRAINAGE, etc.), name, location_description, latitude, longitude, status (WORKING, NEEDS_ATTENTION, NOT_WORKING, RECURRING_FAILURE), risk_level, failure_count, is_recurring_flag
issues	id (PK), asset_id (FK), ward_id (FK), department_id (FK), assigned_to_user_id (FK)	detection_source (CITIZEN_REPORT, SCHEDULED_INSPECTION, COMMUNITY_OBSERVATION), title, description, priority (CRITICAL, HIGH, MEDIUM, LOW), priority_score, priority_rationale, status (11 States), target_completion_date, before_photo_url, after_photo_url, escalation_level
issue_status_history	id (PK), issue_id (FK), changed_by_user_id (FK)	old_status, new_status, notes, created_at (Timestamped immutable audit log of every workflow state change)
inspections	id (PK), ward_id (FK), assigned_to_user_id (FK)	schedule_date, status (SCHEDULED, IN_PROGRESS, COMPLETED), total_assets_checked, issues_detected_count
inspection_items	id (PK), inspection_id (FK), asset_id (FK), created_issue_id (FK)	result (PENDING, WORKING, NEEDS_ATTENTION, NOT_WORKING), notes, photo_url, inspected_at

TABLE NAME	PRIMARY KEY & FOREIGN KEYS	CORE ATTRIBUTES & CHECK CONSTRAINTS
feedback	id (PK), issue_id (FK, UNIQUE), citizen_id (FK)	is_resolved_confirmed (0 or 1), rating (1 to 5), comments, feedback_source, submitted_at
notifications	id (PK), user_id (FK), issue_id (FK)	title, message, type (STATUS_CHANGE, ASSIGNMENT, ESCALATION, RECURRING_ALERT), is_read, email_sent
system_config	key (PK)	value, description, updated_at (Key-value store for global runtime parameters)

8.2 Database Optimization Indexes

To guarantee lightning-fast query execution on low-power village server hardware, the following B-Tree indexes are initialized on startup:

```
CREATE INDEX IF NOT EXISTS idx_issues_status ON issues(status);
CREATE INDEX IF NOT EXISTS idx_issues_ward ON issues(ward_id);
CREATE INDEX IF NOT EXISTS idx_issues_department ON issues(department_id);
CREATE INDEX IF NOT EXISTS idx_issues_assigned ON issues(assigned_to_user_id);
CREATE INDEX IF NOT EXISTS idx_issues_asset ON issues(asset_id);
CREATE INDEX IF NOT EXISTS idx_assets_ward ON assets(ward_id);
CREATE INDEX IF NOT EXISTS idx_assets_status ON assets(status);
CREATE INDEX IF NOT EXISTS idx_notifications_user ON notifications(user_id, is_read);
```

9. RESTful API Specification

9.1 Complete Endpoint Directory

METHOD	ENDPOINT ROUTE	AUTH & RBAC	DESCRIPTION & PAYLOAD SUMMARY
GET	/api/health	Public	Returns system health, pilot village status, and server timestamp.
POST	/api/auth/login	Public	Authenticates user via email/password; returns JWT token and user profile.
POST	/api/auth/register	Public	Self-registration for village residents.
POST	/api/auth/switch-role	Public (Demo)	Instantly switches active user persona between Citizen, Field Staff, and Admin for live demonstration audits.
GET	/api/metadata/wards-departments	Public	Returns all wards, department SLA rules, and available field staff.
GET	/api/assets	Public	Lists all registered public infrastructure assets with filter parameters.
POST	/api/assets	Admin	Creates a new public infrastructure asset in master registry.
PATCH	/api/assets/:id	Admin	Updates asset details, risk level, or maintenance frequency.
GET	/api/issues	Public	Retrieves issues filtered by ward, category, priority, or status.
POST	/api/issues	Authenticated	Registers a new issue; triggers Priority Engine & auto-routing.
PATCH	/api/issues/:id/start-work	Staff / Admin	Lineman transitions issue status to IN_PROGRESS upon site arrival.
PATCH	/api/issues/:id/resolve	Staff / Admin	Lineman submits resolution notes & after-repair photo; status → RESOLVED.
POST	/api/issues/:id/verify	Citizen / Admin	Citizen confirms resolution (status → CLOSED) or rejects (status → REOPENED).
POST	/api/upload	Authenticated	Multipart image file upload; returns accessible public upload URL.

METHOD	ENDPOINT ROUTE	AUTH & RBAC	DESCRIPTION & PAYLOAD SUMMARY
GET	/api/inspections	Authenticated	Fetches scheduled field audit checklists and item defect statuses.
POST	/api/inspections/:id/items:itemId	Staff / Admin	Logs asset inspection result (WORKING, NEEDS_ATTENTION, NOT_WORKING); triggers Proactive Detector if defect found.
GET	/api/admin/analytics	Authenticated	Returns comprehensive KPI suite, 12 chart series, and rule-based insights.
POST	/api/admin/escalations/trigger-check	Admin	Executes batch evaluation of overdue issues; applies Level 1–3 escalations.
GET	/api/notifications	Authenticated	Fetches user notification feed with read/unread flags.

10. User Personas & Frontend Interaction Modules

10.1 Multi-Persona Operating Model

GramSeva is structured around four distinct operational personas, each provided with an interface optimized for their specific workflow and technical literacy level:

Persona 1: Gram Panchayat Administrator (PDO - K. Shivakumar)

- **Full Operational Command:** Real-time overview of village infrastructure health, open work orders, and field crew deployment.
- **Escalation Center:** Instant visibility into overdue SLA tickets, with 1-click trigger to execute automated escalation checks.
- **Chronic Asset Analysis:** Dedicated view of assets failing ≥ 3 times, with administrative recommendations for permanent replacement.
- **Interactive Village Map:** SVG visual map of Honnur displaying ward boundaries, landmarks, and color-coded asset markers.

Persona 2: Village Citizen (E.g., Prajwal Patel, Lakshmi Devi)

- **Simplified Bilingual Filing:** Multilingual interface toggleable between **Kannada (ಕನ್ನಡ)** and English.
- **Landmark-Driven Location:** Pre-populated dropdowns of familiar village landmarks (PHC, High School, Marigudi Temple, Bus Stop).
- **Transparent Lifecycle Tracker:** 5-step visual stepper tracking issue progress from Reported to Closed.
- **Citizen Verification Modal:** Direct empowerment to confirm whether repairs were actually executed, rate quality (1–5 stars), or reopen defective fixes.

Persona 3: Field Staff / Lineman (E.g., Ramesh Kumar, Suresh Gowda)

- **Mobile Task Runner:** Clean, high-contrast mobile checklist for scheduled ward inspection rounds.
- **One-Tap Defect Capture:** Immediate defect marking with camera integration for photo proof.
- **Active Work Order Deck:** Clear list of assigned repairs sorted by SLA priority urgency.
- **Before/After Evidence Chain:** Mandatory requirement to upload photo proof of completed work before a ticket can be moved to Resolved.

Persona 4: Community Observer / Ward Committee

- **Public Transparency Board:** Open dashboard accessible to Ward Committee members, SHG federations, and Asha workers.
- **Reliability Scorecards:** Transparent disclosure of ward-wise resolution rates and department SLA compliance.

11. Deployment, Verification & Future Roadmap

11.1 Local Installation & Execution Guide

GramSeva is pre-configured for frictionless local execution across Windows, macOS, and Linux environments without requiring external database servers or complex cloud dependencies:

```
# 1. Clone repository & install dependencies
git clone <repository-url>
cd leap
npm install
npm install --prefix client

# 2. Database Auto-Initialization
# The backend automatically initializes schema.sql and seeds Honnur GP baseline data on first run.
# To manually re-seed the test environment at any time:
npm run seed

# 3. Start Full-Stack Development Environment
# Concurrently launches Express Backend (Port 5000) and Vite Frontend (Port 3000)
npm run dev

# 4. Access Platform
# Web Portal: http://localhost:3000
# API Health: http://localhost:5000/api/health
```

11.2 Automated Verification & Integration Test Suite

GramSeva includes an end-to-end integration test harness (server/test_workflow.js) that exercises the primary user journey programmatically:

```
# Execute the primary end-to-end user journey test
npm test

Integration Test Sequence Executed:
[STEP 1] Asset Registry: Verifies asset SL-047 (Streetlight near Primary School).
[STEP 2] Scheduled Audit: Lineman Ramesh marks SL-047 as NOT_WORKING with defect note.
[STEP 3] Proactive Detection: Auto-generates Issue GS-2026-1001 with CRITICAL priority (Score: 85).
[STEP 4] Work In Progress: Lineman arrives on site and updates status to IN_PROGRESS.
[STEP 5] Work Completed: Lineman uploads After-Repair photo and marks RESOLVED.
[STEP 6] Citizen Verification: Resident Mahadevappa confirms repair with 5-star rating -> CLOSED.
[STEP 7] Chronic Asset Engine: Simulates repeat failures; verifies escalation to RECURRING_FAILURE.
[STEP 8] Escalation Engine: Simulates overdue SLA breach; verifies auto-escalation to PDO & Taluk EO.
```

11.3 Future Technical Roadmap for Karnataka-wide Scaling

- **IoT Telemetry Integration:** Integrating ultrasonic water level sensors on overhead reservoirs and smart energy sub-meters on borewell pump starters to trigger auto-tickets before manual inspections.
- **Conversational WhatsApp/IVRS Bot:** Introducing low-bandwidth WhatsApp chatbot reporting in spoken Kannada with voice note transcription for non-literate residents.
- **GIS & Bhoomi Integration:** Linking asset coordinates with Karnataka Revenue Department (Bhoomi) parcel boundaries and RDPR e-Swathu assets registry.
- **Taluk-Level Multi-Tenant Aggregation:** Expanding database tenancy to support all 300+ Gram Panchayats across Mandya District with a unified Taluk Panchayat administrative roll-up.

Document Verification Sign-off

Project: GramSeva (ಗ್ರಾಮ ಸೇವಾ) • **Pilot:** Honnur Gram Panchayat, Karnataka

Technical Lead: Prajwal P. (Civic-Tech Systems Architect) • **Reviewed:** September 2026

Certified for production pilot deployment and institutional presentation to Karnataka Department of Rural Development and Panchayati Raj (RDPR).